Analyzing the Impacts of Pets on Household Wellbeing

Hanna Suzuki
Bedford High School, MA, Hanna.S.Suzuki@gmail.com

Introduction

- Households invest heavily in pets (time and $)
  - 2024 (US): 71% of households (94M) owned ≥ 1 pet; $152B spent
  - Time cost: 44 min/day average pet care
    - Similar to grocery trips (46 m) and food prep (53 m)
- 97% of owners say pets are part of the family
- Research gap: pets are understudied in econ

Objectives

- Build an econ model of household pet-ownership choices
  - Core principles: Utility max, budget + time constraints, opportunity cost, diminishing marginal utility
  - Theory link: fertility/child-demand models (pets ↔ children analogy)
- Analyze cost-benefit & comparative statics
- Visualize utility landscape & param relations
- Enable quantitative reasoning for current & prospective owners

The Proposed Model

- Household chooses pet quantity (n), caregiving intensity (c), and consumption (standard of living; s) to maximize welfare (utility; U):

$$U(s,n,c) = s + \underbrace{\alpha \ln(\beta n c^{\gamma})}_{\text{companionship from pets}}$$

- Budget-time constraint: $w = s + \underbrace{(wt + pc + k)}_{\text{cost of a pet}} n$
- Reduced form:

$$U(n,c) = w - wtn - pcn - kn + \alpha \ln(\beta) + \alpha \ln(n) + \alpha\gamma \ln(c)$$

- First-order conditions :

$$\partial U/\partial n = \alpha/n - (wt + pc + k) = 0$$
$$\partial U/\partial c = \alpha\gamma/c - pn = 0$$

- Optimal choices:

$$n^* = \frac{\alpha(1-\gamma)}{wt+k} \qquad c^* = \frac{\gamma(wt+k)}{p(1-\gamma)} \qquad s^* = w - \alpha$$

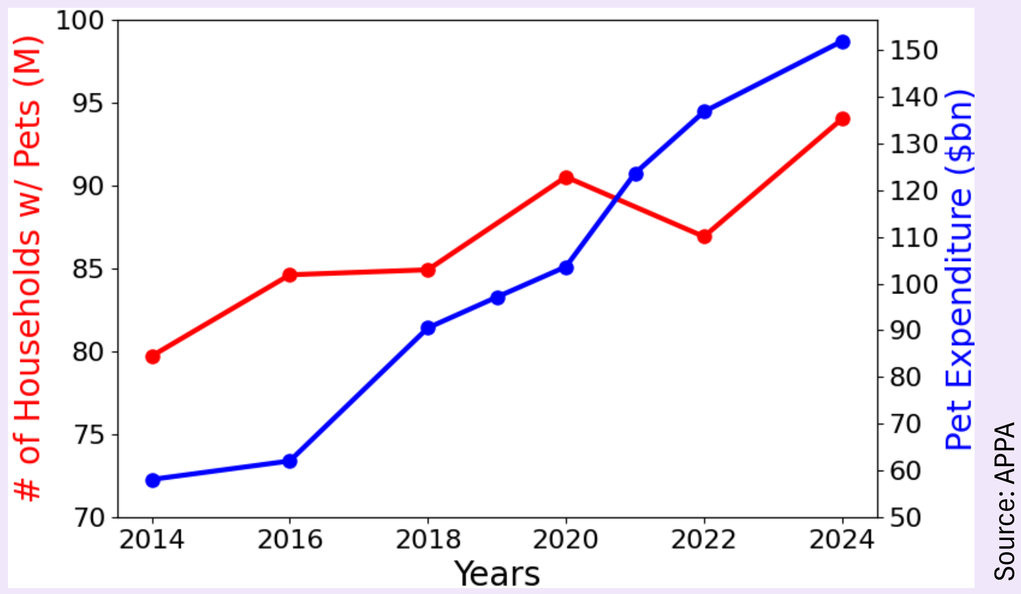
- Exogenous parameters:

α: preference for pets	t: per-pet caregiving time
γ: caregiving efficiency	p: pet-care goods price
w: wage	k : per-pet ownership overhead

Comparative Statics

- Hypotheses: How param shifts affect household choices (cause → effect)

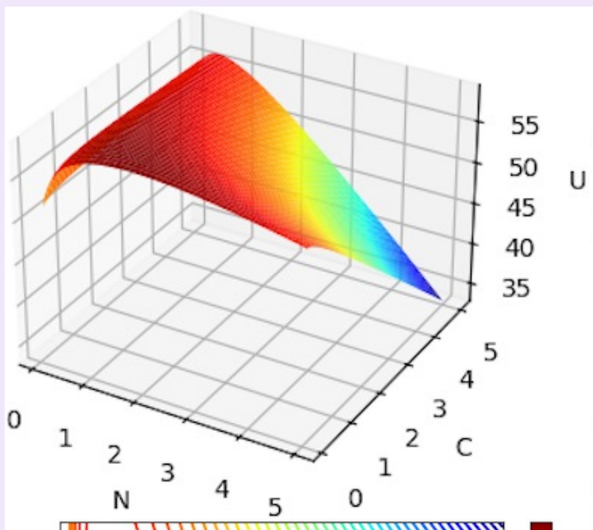
	α	γ	w	p	k	t
Pet quantity: n^*	+	–	–	0	–	–
Care intensity: c^*	0	+	+	–	+	+
Standard of living: s^*	–	0	+	0	0	0
Pet expenditure: pc^*n^*	+	+	0	0	0	0
Total care time: tn^*	+	–	–	0	–	+

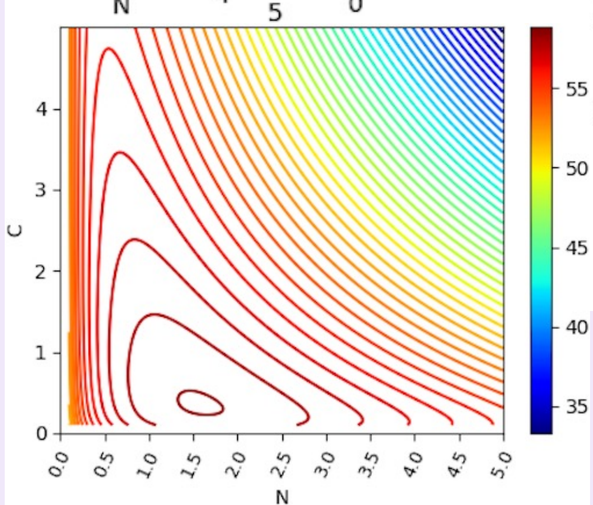


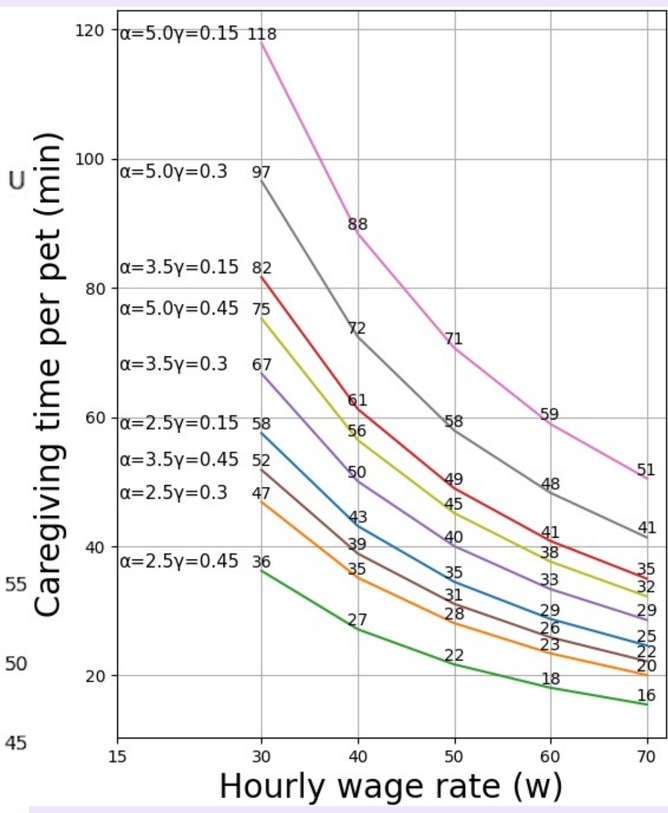
- 2020-2021 pet quantity (n^*) spike (Covid-19)
  - $t \downarrow \rightarrow n \uparrow$: More daily avail time (lockdown)
  - $\alpha \uparrow \rightarrow n \uparrow$: Stronger preference (loneliness, anxiety)
- 2022-2023 pet quantity (n^*) dip
  - $t \uparrow \rightarrow n \downarrow$: Less daily avail time (return-to-office)
  - $\alpha \downarrow \rightarrow n \downarrow$: Lower pref (more leisure substitutes)
  - $k \uparrow \rightarrow n \downarrow$: Inflation raising overhead (e.g. insurance)
- Consistent pet expenditure (pc^*n) rise
  - $t \uparrow \rightarrow c \uparrow \rightarrow pcn \uparrow$: Lower daily avail time
  - $\gamma \uparrow \rightarrow pcn \uparrow$: Higher care efficiency thru "pet tech"
  - $\alpha \uparrow \rightarrow pcn \uparrow$: GenZ pet ownership rising rapidly

Computational Experiments

- Calibrated with 2024 US data







$$n^* = \frac{\alpha(1-\gamma)}{wt+k} = 1$$